Aortic Dissection

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Section I: Scenario Demographics

Scenario Title:	Aortic Dissection
Date of Development:	(10/01/2016)
Target Learning Group:	☐ Juniors (PGY 1 – 2) ☐ Seniors (PGY ≥ 3) ☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Martin Kuuskne
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To recognize and manage a critically ill patient with aortic dissection and its potential cardiac complications.
CRM Objectives:	1) Mobilize appropriate human resources in the work-up and disposition of aortic dissection. 2) Maintain and actively verbalize a wide differential diagnosis for the critically ill patient with chest pain. 3) Set priorities dynamically as the patient's status changes.
Medical Objectives:	1) Appropriately manage hypertension in the setting of aortic dissection with blood pressure and heart rate targets. 2) Recognize EKG manifestations of RCA involvement and resultant ischemia in the setting of aortic dissection. 3) Appropriately apply the ACLS algorithm for unstable bradycardia and asystole.

Case Summary: Brief Summary of Case Progression and Major Events
A 66 year old female with a history of smoking, HTN and T2DM presents with syncope while walking her dog. She complains of retrosternal chest pain radiating to her jaw. She will become increasingly bradycardic and hypotensive, requiring the team to mobilize resources in order to facilitate diagnosis and management of an aortic dissection.

References
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.
http://lifeinthefastlane.com/ccc/acute-aortic-dissection/

Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case
You are working the day shift at a tertiary-care hospital. A 66-year old female is being wheeled into the resuscitation bay with a history of as syncopal episode. No family members or friends are present with the patient.

B. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient	*Select most important dimension(s)*	☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates	**Brief Description of Role**		
RN	To assist if asked, provide relevant history and current status of patient. Patient was walking her dog and had a syncopal episode witnessed by bystanders who called EMS.		

C. Required Monitors		
☒ EKG Leads/Wires	☒ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:

D. Required Equipment		
☒ Gloves	☒ Nasal Prongs	☐ Scalpel
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☒ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit
☐ Blood Products	☒ ET Tubes	☐ Other:
☐ Intraosseous Set-up	☐ LMA	☐ Other:

E. Moulage
Mannequin dressed in female outdoor clothing

F. Approximate Timing					
Set-Up:	10 min	Scenario:	10 min	Debriefing:	15 min

Section V: Patient Data and Baseline State

A. Patient Profile and History			
Patient Name: Dorothy Williams		Age: 66	Weight: 50 kg
Gender: ☐ M ☒ F		Code Status: Full	
Chief Complaint: Syncope			
History of Presenting Illness: Walking the dog outside and suddenly fainted, seen by passerby who called ambulance.			
Past Medical History:	HTN	Medications:	Amlodipine
	T2DM		Metformin
	No prev MI/CVA/PE		ASA
Allergies: NKDA			
Social History: 30 pack year smoker			
Family History: None contributory			

Review of Systems:				
	CNS:	None		
	HEENT:	None		
	CVS:	Retrosternal chest pain radiating to right jaw, constant, sharp, x30 mins		
	RESP:	None (no SOB, no hemoptysis)		
	GI:	None		
	GU:	None		
	MSK:	None	INT:	None

B. Baseline Simulator State and Physical Exam			
☒ No Monitor Display	☐ Monitor On, no data displayed	☐ Monitor on Standard Display	
HR: 110/min	BP: 210/100 (160/80 left arm only if asked)	RR: 20/min	O_2Sat: 99%
Rhythm: Sinus tach	T: 36.5°C	Glucose: 5.0 mmol/L	GCS: 15 (E4 V5 M6)
General Status: No acute distress, no respiratory distress			

CNS:	Normal		
HEENT:	Normal		
CVS:	Diastolic III/VI murmur		
RESP:	Bi-basilar crackles		
ABDO:	Normal		
GU:	Normal		
MSK:	Normal	INT:	Normal

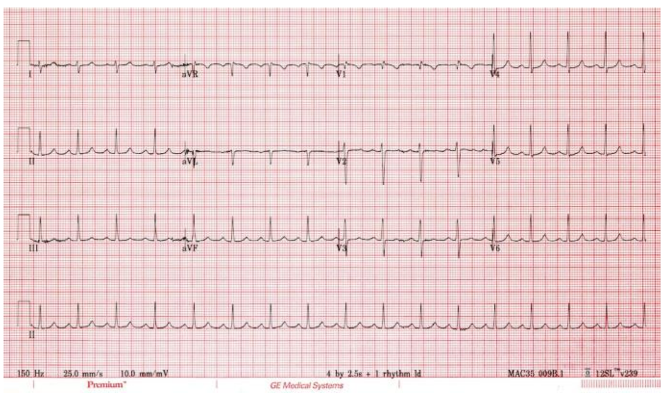
Section VI: Scenario Progression

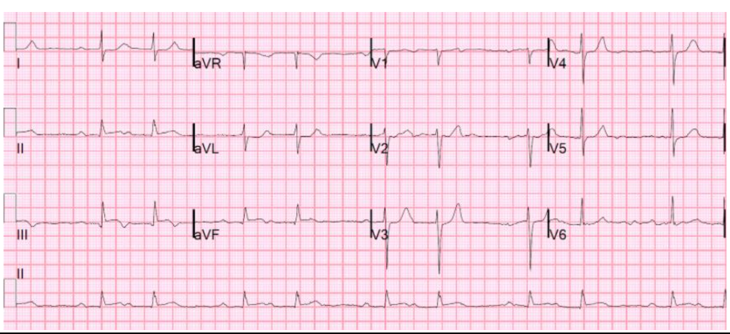
Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus tach HR: 110/min BP: 210/100 (R) 160/80 (L) only if asked RR: 20/min O_2Sat: 99% T: 36.5°C	Alert. Complaining of chest pain	Learner Actions -☐ Full set of vitals, monitors -☐ ECG -☐ CXR -☐ Start 2 Large Bore IV lines -☐ IV Labetolol OR -☐ IV Esmolol + Nitroprusside -☐ Orders Emergent CT chest	Modifiers *Changes to patient condition based on learner action* - Give negative inotrope (HR→85, BP→150/96) - Give vasodilator (HR→85, BP→135/80) - Give NTG SL (BP→200/100 then increases) Triggers *For progression to next state* - Give heparin → **4. Bradycardia-Asystolic arrest** - 4 minutes → **2. Increasing Chest Pain**
2. Increasing Chest Pain HR→50 BP→90/60 and decreasing	Complaining of ***severe*** chest pain	Learner Actions -☐ Repeat EKG (Mobitz I, inferior STE) -☐ Discontinue any prior β-blockers or vasodilators. -☐ Give IV fluids -☐ Atropine IV -☐ Call for CT chest (if not yet) -☐ Bedside U/S for PCE	Modifiers - Give IV fluids → BP 100/65 - Give atropine →no change Triggers - 3 minutes → **3. Unstable Bradycardia**
3. Unstable Bradycardia HR→40/min BP→70/30 RR→12/min	Acute distress	Learner Actions -☐ Initiate intubation -☐ Atropine IV -☐ Initiate transcutaneous pacing -☐ Consider IV inotropic/chronotropic agents -☐ Call ICU -☐ Bedside ultrasound for PCE -☐ Call for TEE -☐ Consult Cardiac surgery	Modifiers - Give atropine → no change - Give IV chronotropic agents → BP 90/50 - Transcutaneous pacing → BP 90/50 Triggers **- END SCENARIO PRN with TEE or consultant arrival**
4. Bradycardia-Asystolic arrest Rhythm→asystole HR→0/min BP→0/0 RR→0/min O_2Sat→0%	Non-responsive	Learner Actions -☐ High Quality CPR -☐ Epinephrine 1amp q3-5min -☐ ACLS -☐ Crash Intubation -☐ Monitor capnography	Modifiers Triggers - 1st round of CPR and epi given → **3. Unstable Bradycardia** -Total scenario >10 minutes → **END SCENARIO**

Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 137	K: 4.9	Cl: 101	HCO_3: 24	BUN:	Cr:	Glu: 5.0
Ca:	Mg:	PO_4:	Albumin:			
VBG	pH: 7.23	PCO_2: 30	PO_2:	HCO_3: 25	Lactate:2.9	
WBC: 9.0	Hg: 140	Hct:	Plt: 360			
Troponin: 358						

Images (ECGs, CXRs, etc.)





Retrieved from: https://radiopaedia.org/articles/aortic-dissection

Ultrasound Video Files (if applicable)

Section VIII: Debriefing Guide

General Debriefing Plan			
☐ Individual	☒ Group	☐ With Video	☒ Without Video

Objectives	
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Sample Questions for Debriefing
1. At what point did you suspect this was a dissection? How common is syncope a presentation of dissection? (9%) AMI?(7%) Pulmonary Embolism? (10%) 2. What other items were on your differential? 3. What is the most time efficient way to rule out items on the differential for chest pain? 4. What is the best test that can be performed most EDs to rule in a dissection in the unstable patient? 5. How did your management priorities change for this patient with the change in vital signs? 6. Did the team feel like a cohesive unit today? Were there any challenges to your leadership or communication?

Key Moments
Recognition of aortic dissection
Change in management when patient became hypotensive
Initiation of transcutaneous pacing when patient became bradycardic